

Harrison J. Souchereau

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Education

Yale University, Ph.D in Astronomy (ongoing), advised by Jeffrey Kenney	2021 -
Yale University, MS and M.Phil in Astronomy	2021-2024
Saint Mary's University, BS in Astrophysics	2016 - 2020
• Graduated Magna Cum Laude • Honors Thesis Title: <i>The Performance of Surface Brightness Profile Extraction</i>	

Professional Experience

Pre-doctoral Fellow, Flatiron Institute, Simons Foundation	02/2024 - 07/2024
• Center for Computational Astrophysics, advised by Dr Stephanie Tonnesen • 162 Fifth Avenue, New York NY, 10010	
McDougal Fellow, Office of GSAS Graduate Student Professional Development	2025-
Senator, Yale Graduate and Professional Student Senate (GPSS)	2024 -
• Elected by the graduate student body.	
Programmer / Data Analyst, Saint Mary's University	05/2018 - 07/2021
Programmer, Saint Mary's University	10/2019 - 06/2020
• Designed a classification workflow in PYTHON to identify faint features of galaxy mergers in the SDSS sample.	

First-Author Publications

3. **Souchereau H**, Tonnesen S, Zhu J, et al., *Up, Up, and Away? Quantifying ISM Fallback using Ram Pressure Stripping Simulations* (submitted)
2. **Souchereau H**, Kenney J, Jachym P, et al., *ALMA-JELLY I: High Resolution CO(2-1) Observations of Ongoing Ram Pressure Stripping in NGC 4858 Reveal Asymmetrical Gas Tail Formation and Fallback* (Published in ApJ, DOI: 10.3847/1538-4357/adde47).
1. **Souchereau H**, Damjanov I, Sawicki M, et al., *GALPRIME: Automated Quality-Assessed 1-D Light Profile Extraction for Modern Galaxy Surveys* (submitted)

Other Publications

4. Williams D, Damjanov I, Sawicki M, **Souchereau H**, et al., *The Growth of Galaxy Stellar Haloes Over $0.2 \leq z \leq 1.1$* , Submitted to ApJ
3. Keim, M., van Dokkum, P., Shen, Z., **Souchereau, H.**, et al., *Kinematic Confirmation of a Remarkable Linear Trail of Galaxies in the NGC 1052 Field, Consistent with Formation in a High-Speed Bullet Dwarf Collision*
2. H. Domínguez Sánchez, et al. (incl. **Souchereau, H.**) *Identification of tidal features in deep optical galaxy images with Convolutional Neural Networks*, MNRAS., doi:10.1093/mnras/stad750 (2023)
1. Damjanov, I., Geller, M. J., Utsumi, Y., Sohn, J., & **Souchereau, H.**, *Quiescent Galaxy Size and Spectroscopic Evolution: Combining HSC Imaging and Hectospec Spectroscopy*, The Astrophysical Journal 872.1 (2019): 91.

Grants, Scholarships, and Awards

8. NRAO Student Observing Support (SOS) Program Grant	2024-2025
7. Yale GSA Conference Travel Fund Grant (\$800)	2024
6. Gold Award , The Duke of Edinburgh's International Award	2021
5. SMU Faculty of Science Conference Travel Grant, (\$500)	2019
4. SMU Academic Achievement Award, (\$3000)	2018, 2019
3. VPAR International Mobility Award	2017
2. Dalhousie University International Baccalaureate Entrance Scholarship, (\$20000)	2015
1. International Baccalaureate <i>Theory of Knowledge</i> Subject Award	2015

Outreach

2. Yale Grad Fair Recruiter Joint NSBP/NSHP Conference	
• Attended the Annual conference in San Jose, CA.	Fall 2025
• Attended the Annual conference in Houston, TX.	Fall 2024
1. Yale Grad Fair Recruiter , National Society of Black Physicists (NSBP)	
• Attended the NSBP Annual conference in Knoxville, TN.	Fall 2023
• Attended the NSBP Annual conference in Charlottesville, VA.	Fall 2022

Science Outreach and Community Involvement

6. Director , Yale Science Communications	Fall 2023 -
5. Volunteer and Photographer , Astronomy on Tap New Haven	2021-2023
4. Mentor , Yale Astronomy <i>AstroSibs</i> Program	2021-2023
3. Presenter , Leitner Family Observatory and Planetarium	2021-
2. President , SMU Astronomy and Physics Society	2019-2020
1. Vice-President , SMU Astronomy and Physics Society	2018-2019

Teaching

2. Teaching Fellow , Yale University	
<i>Galaxies</i>	Fall 2023, Fall 2024
<i>Origins and the Search for Life in the Universe</i>	Spring 2023
<i>Frontiers and Controversies in Astrophysics</i>	Fall 2022
<i>Planets and Stars</i>	Summer 2022
<i>Introduction to Astronomical Observing</i>	Fall 2021
<i>Galaxies and the Universe</i>	Summer 2021/2022, Spring 2022
1. Teaching Assistant , Saint Mary's University	
<i>The Physics of Stars</i>	Fall 2019

Selected Attended Workshops

2. YCRC Machine Learning and Big Data Workshop	03/2023
- Hands on experience using SPARK for handling enormous datasets.	
- Lectures on machine learning, and using TENSORFLOW for deep learning.	
1. Canadian Astroparticle Physics Summer School (CAPSS)	05/2019
- Offered by Queens University in Hamilton, Ontario	
- 1-week long summer school with lectures on astroparticle science, and a tour of the underground research facility SNOLAB in Sudbury, Ontario.	

Observing Experience

- Keck II (10m), KWI - W.M. Keck Observatory, Hawaii: Awarded 2.5 nights
- Keck I (10m), LRIS - W.M. Keck Observatory, Hawaii: Assisted 4 nights
- Burke Gaffney Observatory, Saint Mary's University, Halifax: Awarded 3 nights

Presentations and Talks

9. *Inner Tail Gas Asymmetries and Fallback in the Jellyfish Galaxy NGC 4858*
- Presented at the 246th Meeting of the American Astronomical Society in Anchorage, AK.
8. *Ram Pressure Recycling: Inner Tail Gas Asymmetries and Fallback in the Jellyfish Galaxy NGC 4858* 2024
- Presented at the Jellyfish Galaxies and ram-pressure stripping workshop at the Paris Observatory
- Presented at the research seminar at the Astronomical Institute of the Czech Academy of Sciences
7. *Efficiently Exploring the Ram Pressure Space* 06/2024
- Presented at the Center for Computational Astrophysics Pre-Doctoral Symposium
6. *NGC 4858: Rotation Dancing with Ram Pressure* 03/2023
- Presented at the Yale Astronomy Professional Seminar.
5. *The End of a Galaxy's Life* 03/2023
- Presented as part of the Yale Science Communication Society's textbfScience in the News talks.
- Presented at 4 libraries in the CT area.
- Talk was part of *From Guts to Galaxies: Evolution on Many Scales*
4. *Broad Slit Spectroscopy of Ultra-Diffuse Galaxies* 04/2022
- Presented at the Yale Astronomy Professional Seminar.
3. *Surface Brightness Profiles and Galactic Evolution* 11/2019
- Presented at the Canadian Undergraduate Physics Conference (CUPC),
- Hosted by McGill University, Montreal QC.
2. *Surface Brightness Profile Extraction and Analysis for Big Data* 09/2019
- Presented at the Symposium for Undergraduate Student Research, Saint Mary's University, Halifax NS.
1. *Examining Factors in Galactic Evolution Using Surface Brightness Profiles* 09/2018, 02/2019
- Presented at the Atlantic Universities Physics and Astronomy Conference (AUPAC), Acadia University, Wolfville NS.
- Presented at the Symposium for Undergraduate Student Research, Saint Mary's University, Halifax NS.

Professional Societies

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| Yale Graduate and Professional Student Senate (GPSS) | 2024- |
| Yale Graduate Student Assembly (GSA) | 2024-2025 |
| American Astronomical Society (AAS) | Joined 2021 |
| Saint Mary's University Debate Society (SMDS) | 2019-2020 |
| Saint Mary's University Astronomy and Physics Society (SMUAPS) | 2017-2020 |

Technologies

Languages: Python (proficient), Rust, C, Java, SQL, Bash

Technologies: Slurm, Anaconda, CASA, SaoImageDS9 / CARTA
TouchDesigner